

Python For Loop Problems for Practice and Teaching

Beginner Level For Loop Problems

1. Print Numbers from 1 to 10

```
for i in range(1, 11):  
    print(i)
```

1. Print Even Numbers from 1 to 20

```
for i in range(2, 21, 2):  
    print(i)
```

1. Sum of First 10 Natural Numbers

```
total = 0  
for i in range(1, 11):  
    total += i  
print("Sum:", total)
```

1. Print Multiplication Table of a Number

```
num = 5  
for i in range(1, 11):  
    print(f"{num} x {i} = {num * i}")
```

1. Print Each Character of a String

```
text = "Python"  
for char in text:  
    print(char)
```

5 Intermediate Level For Loop Problems

1. Count Vowels in a String

```
text = "hello world"  
count = 0  
for char in text:
```

```
    if char.lower() in 'aeiou':  
        count += 1  
print("Vowel Count:", count)
```

1. Print List in Reverse

```
numbers = [1, 2, 3, 4, 5]  
for i in range(len(numbers)-1, -1, -1):  
    print(numbers[i])
```

1. Find the Largest Number in a List

```
numbers = [12, 45, 2, 67, 33]  
max_num = numbers[0]  
for num in numbers:  
    if num > max_num:  
        max_num = num  
print("Largest:", max_num)
```

1. Print Pattern

```
*  
**  
***  
****  
*****
```

```
for i in range(1, 6):  
    print("*" * i)
```

1. Check if a Number is Prime

```
num = 29  
is_prime = True  
for i in range(2, int(num**0.5) + 1):  
    if num % i == 0:  
        is_prime = False  
        break  
print("Prime" if is_prime else "Not Prime")
```

Bonus Problem (Real-life Context)

1. Calculate Total Bill from Item Prices

```
prices = [120, 250, 310, 89]
total = 0
for price in prices:
    total += price
print("Total Bill:", total)
```